MINLP Modeling

Bjarne Jehn

July 17, 2026

One Level

Runtime

As a first approach we model the runtime complexity for only one level.
In the following we will use capital letters for constants and bold letters for variables.

constants	meaning
T_R	time to check presence of a bit in vector R of level i
T_{SL}	time to lookup a k-mer in vector S on level i
T_H	time to lookup a k-mer on level i
$P_{K-empiric}(m,t)$	empirically estimated value for a kmer with a minimizer of length m to have a frequency less than t
$\mathcal{M}$	Set of all minimizer length to be considered
$\mathcal{T}$	Set of thresholds to be considered
variables	
$\mathbf{p_K}^{L_i}$	probability of a kmer belonging to level i
$\mathbf{sm}_m$	binary selector for minimizer length m
$\mathbf{st}_t$	binary selector for threshold of t

For one level the lookup of a query with minimizer m of frequency f(m) would consist of checking the bitvector R in order to determin whether $1 \leq f(m) \leq threshold$. This will be the case for $\mathbf{p_K}$ of the k-mers, which will then be followed by a lookup in S. If m does not appear at all or has a $f(m) > threshold$, a lookup in the hashtable will follow. The lookup time for one level can thus be modeled as:

$$T_R + {T_{SL}}^{L_1} * \mathbf{p_K} + T_H * (1 - \mathbf{p_K})$$

Let $\mathcal{M}$ be the Set of all minimizer length and $\mathcal{T}$ the set of all thresholds to be considered.
In order to determine m and t we introduce binary selector variables $\mathbf{sm}_m$, $\mathbf{st}_t$ for each possible $m \in \mathcal{M}$ and $t \in \mathcal{T}$.

$$\mathbf{sm}_m = \begin{cases} 1 & \text{minimizer length is } m \\ 0 & \text{otherwise.} \end{cases}$$

$$\mathbf{st}_t = \begin{cases} 1 & \text{threshold is } t \\ 0 & \text{otherwise.} \end{cases}$$

For a fixed minimizer length m and threshold t let $P_{K-empiric}(m,t)$ be the empirically determined percentage of k-mers that have a minimizer with $frequency < t$.

We can then model $\mathbf{p_K}$ with:

$$\mathbf{p_K} = \sum_{m \in \mathcal{M}} \left(\mathbf{sm}_m \sum_{t \in \mathcal{T}} P_{K-empiric}(m,t) * \mathbf{st}_t \right)$$

$$\implies \mathbf{p_K} - \sum_{m \in \mathcal{M}} \left(\mathbf{sm}_m \sum_{t \in \mathcal{T}} P_{K-empiric}(m,t) * \mathbf{st}_t \right) = 0$$

In order for this to work, the selectors for m and t must be binary and mutually exclusive:

$$\mathbf{sm}_m \in 0,1 \mid \forall m \in \mathcal{M}$$
$$\mathbf{st}_t \in 0,1 \mid \forall t \in \mathcal{T}$$
$$\sum_{m \in \mathcal{M}} \mathbf{sm}_m = 1$$
$$\sum_{t \in \mathcal{T}} \mathbf{st}_t = 1$$

We can then model $\mathbf{m}$ and $\mathbf{t}$ as:

$$\mathbf{m} = \sum_{m \in \mathcal{M}} \mathbf{sm}_m * m$$
$$\mathbf{t} = \sum_{t \in \mathcal{T}} \mathbf{st}_t * t$$

The above results in the following MINLP (Mixed Integer Non Linear Program):

MINIMIZE

$$T_R + {T_{SL}}^{L_1} * \mathbf{p_K} + T_H * (1 - \mathbf{p_K})$$

SUBJECT TO

$$\mathbf{p_K} - \sum_{m \in M} \left(\mathbf{sm}_m \sum_{t \in T} P_{K-empiric}(m,t) * \mathbf{st}_t \right) = 0$$
$$\mathbf{sm}_m \in 0,1 \mid \forall m \in M$$
$$\mathbf{st}_t \in 0,1 \mid \forall t \in T$$
$$\sum_{m \in M} \mathbf{sm}_m = 1$$
$$\sum_{t \in T} \mathbf{st}_t = 1$$
$$\mathbf{m} = \sum_{m \in \mathcal{M}} \mathbf{sm}_m * m$$
$$\mathbf{t} = \sum_{t \in \mathcal{T}} \mathbf{st}_t * t$$

For now we will assume that $T_{SL} \approx 20ns$ and $T_H \approx 40ns$.
Solving this MINLP will result in a Solution where the $\mathbf{sm}_m$, $\mathbf{st}_t$ corresponding to the largest $m \in \mathcal{M}$, $t \in \mathcal{T}$ will be set to one.
This is as expected, since this leads to the highest possible $\mathbf{p_K}$. This maximizes the k-mers found in the vector S and thus minimizes the lookups in the (slower) Hashtable.

However this also results in a high space consumption, because explain why

In order to take that into account, we have to add a constraint modeling the space consumption.

Space Consumption

The space consumption of the Data Structure used by RSHash can be split into that of the Vectors R, O and S as well as the hashtable H.

Bitvectors R and S

R and S can likely be disregarded when modeling the space consumption. explain why

Offsetvector O

O contains one entry of size $log|T|$ for the offset of every minimizer with frequency $\leq t$ in T.

$$s_O = log_2|T| * |O| = log_2|T| * \sum_{m=1}^{t} \mathbf{m} * (\text{number of minimizers with frequenzy m})$$

Every minimizer of length $\mathbf{m}$ covers an average of $\frac{k-\mathbf{m}+2}{2}$ consecutive k-mers. We can thus model the bits per k-mer in O as:

$$bpk_O = \frac{2}{k - \mathbf{m} + 2} * \mathbf{p_K} * log|T|$$

Hashtable H

Several Levels

Runtime

minimize: nicht linear!

$$
\begin{aligned}
{T_R}^{L_1} \quad & +\mathbf{p_K}^{L_1} * {T_{SL}}^{L_1} \\
& +(1-\mathbf{p_K}^{L_1}) * \Bigg({T_R}^{L_2} \quad +\mathbf{p_K}^{L_2} * {T_{SL}}^{L_2} \\
& \qquad\qquad +(1-\mathbf{p_K}^{L_2}) * \Big({T_R}^{L_3} \quad +\mathbf{p_K}^{L_3} * {T_{SL}}^{L_3} \\
& \qquad\qquad\qquad\qquad +(1-\mathbf{p_K}^{L_3}) * T_H \Big) \Bigg)
\end{aligned}
$$

TODO: linearise

variable	meaning
$\mathbf{p_K}^{L_i}$	probability of a kmer belonging to level i
${T_R}^{L_i}$	time to check presence of a bit in vector R of level i
$t_l^{L_i}$	time to lookup a k-mer on level i

We can model $\mathbf{p_K}$ for multiple levels with:

$$
\begin{aligned}
\mathbf{p_K}^{L_i} &= \sum_{m \in M} \left(\mathbf{sm}_m^{L_i} \sum_{t \in T} P_{K-empiric}^{L_i}(m,t) * \mathbf{st}_t^{L_i} \right) \\
\mathbf{p_K}^{L_i} &- \sum_{m \in M} \left(\mathbf{sm}_m^{L_i} \sum_{t \in T} P_{K-empiric}^{L_i}(m,t) * \mathbf{st}_t^{L_i} \right) = 0
\end{aligned}
$$

TODO: linearise

variable	meaning
$\mathbf{sm}_m^{L_i}$	binary selector for minimizer length m in level i
$\mathbf{st}_t^{L_i}$	binary selector for threshold of t in level i
$P_{K-empiric}^{L_i}(m,t)$	empirically estimated value for a kmer with a minimizer of length m to have a frequency less than t
M	Set of all minimizer length to be considered
T	Set of thresholds to be considered

selectors for m and t must be binary amd mutually exclusive:

$$\begin{aligned} \mathbf{sm}_m^{L_i} &\in 0,1 \mid \forall m \in M \\ \mathbf{st}_t^{L_i} &\in 0,1 \mid \forall t \in T \\ \sum_{m \in M} \mathbf{sm}_m^{L_i} &= 1 \\ \sum_{t \in T} \mathbf{st}_t^{L_i} &= 1 \end{aligned}$$

Without further constraints

Space Consumption

Using Elias Fano Coding a Bitvector of length U with n bits set to 1 can be encoded using $n\lceil log\frac{U}{n}\rceil + 3n$.

Let *noam* be the number of appearing minimizers (minimizers with a frequency greater than zero).

Level 1

R - Presence of k-mers:
probably not relevant R^{L_1} has length $4^{sm_m^{L_1}}$ and contains $p_M^{L_1} * noam$ many 1s.

$$\begin{aligned} s_R^{L_1} &= p_M^{L_1} * noam \lceil log \frac{4^{sm_m^{L_1}}}{p_M^{L_1} * noam} \rceil + 3 * p_M^{L_1} * noam \\ &= p_M^{L_1} * noam \left(\lceil log \frac{4^{sm_m^{L_1}}}{p_M^{L_1} * noam} \rceil + 3 \right) \end{aligned}$$

S - Offset Boundaries:
S is a bitvector of length $|O| + 1$ and is **probably not relevant** for the optimization of m and t.

O - Offsets:
O^{L_1} contains one entry of size $log|T|$ for the offset of every minimizer with frequency $\leq t^{L_1}$ in T.

$$s_O = log_2|T| * |O| = log_2|T| * \sum_{m=1}^{t^{L_1}} m * (\text{number of minimizers with frequenzy m})$$

Every minimizer of length m covers an average of $\frac{k-m+2}{2}$ consecutive k-mers.

$$bpk_O^{L_1} = \frac{2}{k - m^{L_1} + 2} * p_K^{L_1} * log|T|$$